2025

Business Report

DATA DRIVEN INSIGHTS USING PYTHON: AUSTO AUTOMOBILES

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2. Business Problem

2.1 Problem Statement

Austo Motor Company is unsure whether its current marketing campaign is effectively reaching the right customers and influencing their vehicle purchase decisions. The lack of clear insights into customer preferences and spending behaviour makes it difficult to optimize marketing efforts.

2.2 Objective

To analyze customer data and understand buying patterns, preferences, and spending behaviour in order to support better marketing decisions and improve overall customer experience.

2.3 Business Opportunity

By identifying key customer segments and understanding what drives their vehicle choices, Austo Motor Company can design more targeted marketing campaigns, improve conversion rates, and increase sales across SUV, Sedan and Hatchback models.

2.4 Problem Definition

The company needs to determine how customer characteristics such as gender, income, profession, family status, and loan behaviour influence vehicle type selection and spending levels, so that marketing strategies can be aligned with actual customer demand.

3. Data Description (Collection & Background)

The dataset contains demographic, financial, and purchase related information of customers who have purchased automobiles from Austo Motor Company. Each variable is described below:

- **Age**: Represents the age of the customer in years.
- **Gender**: Indicates the gender of the customer (Male or Female)
- **Profession**: Describes the customer's occupation category.
- **Marital_status**: Shows the marital status of the customer (Married or Single)
- **Education**: Indicates the highest educational qualification of the customer (Graduate or Post Graduate)
- **No_of_Dependents**: The number of individuals financially dependent on the customer, such as children or elderly family members.
- **Personal_loan**: Indicates whether the customer has taken a personal loan (Yes or No)
- **House_loan**: Indicates whether the customer has an active housing loan (Yes or No)
- **Partner_working**: Indicates whether the customer's partner is employed (Yes or No)
- **Salary**: The monthly or annual income of the customer.
- **Partner_salary**: The income of the customer's partner, where applicable.
- **Total_salary**: The combined income of the customer and their partner.
- **Price**: The purchase price of the automobile.
- **Make**: The type of automobile purchased, categorized as SUV, Sedan, or Hatchback.

4. Data Processing

4.1 Data Shape

The dataset consists of 1,581 records and 14 variables.

4.2. Data Background and content

- The dataset contains 1,581 observations across 14 variables, comprising a mix of numerical and categorical data. Numerical data include age, income related fields, and vehicle price, while categorical variables capture customer demographics, financial status and vehicle type.

#	Column	Non-NULL Count	Dtype
0	Age	1581	Int64
1	Gender	1528	Object
2	Profession	1581	Object
3	Marital_status	1581	Object
4	Education	1581	Object
5	No_of_Dependents	1581	Int64
6	Personal_loan	1581	Object
7	House_loan__	1581	Object
8	Partner_working	1581	Object
9	Salary	1581	Int64
10	Partner_salary	1475	Float64
11	Total_salary	1581	Int64
12	Price	1581	Int64
13	Make	1581	Object

- ➢ Gender & Partner_salary have NULL values.

4.3 Initial Data Preview

A preliminary review of the dataset shows that each record represents an individual customer along with their demographic details, financial information, and vehicle purchase choice.

The first 5 rows are displayed below:

Age	Gender	Profession	Marital_status	Education	No_of_Dependents	Personal_loan	House_loan	Partner_working	Salary	Partner_salary	Total_salary	Price	Make
53	Male	Business	Married	Post Graduate	4	No	No	Yes	99300	70700	170000	61000	SUV
53	Femal	Salaried	Married	Post Graduate	4	Yes	No	Yes	95500	70300	165800	61000	SUV
53	Female	Salaried	Married	Post Graduate	3	No	No	Yes	97300	60700	158000	57000	SUV
53	Female	Salaried	Married	Graduate	2	Yes	No	Yes	72500	70300	142800	61000	SUV
53	Male	Salaried	Married	Post Graduate	3	No	No	Yes	79700	60200	139900	57000	SUV

The last 5 rows are displayed below:

Age	Gender	Profession	Marital_status	Education	No_of_Dependents	Personal_loan	House_loan	Partner_working	Salary	Partner_salary	Total_salary	Price	Make
22	Male	Salaried	Single	Graduate	2	No	Yes	No	33300	0	33300	27000	Hatchback
22	Male	Business	Married	Graduate	4	No	No	No	32000		32000	31000	Hatchback
22	Male	Business	Single	Graduate	2	No	Yes	No	32900	0	32900	30000	Hatchback
22	Male	Business	Married	Graduate	3	Yes	Yes	No	32200		32200	24000	Hatchback
22	Male	Salaried	Married	Graduate	4	No	No	No	31600	0	31600	31000	Hatchback

- The data appears well structured, with clear and meaningful variable values.
- This initial inspection confirms that the dataset is reliable and ready for further data cleaning and exploratory analysis.

4.4. Checking the Statistical Summary of the data

index	Age	No_of_Dependents	Salary	Partner_salary	Total_salary	Price
count	1581.0	1581.0	1581.0	1581.0	1581.0	1581.0
mean	31.922201138519924	2.4579380139152436	60392.220113851996	19233.776091081592	79625.99620493359	35597.722960151805
std	8.425978166558165	0.9434829353622931	14674.825043862535	19670.39117133602	25545.8577676707	13633.636545130645
min	22.0	0.0	30000.0	0.0	30000.0	18000.0
25%	25.0	2.0	51900.0	0.0	60500.0	25000.0
50%	29.0	2.0	59500.0	25100.0	78000.0	31000.0
75%	38.0	3.0	71800.0	38100.0	95900.0	47000.0
max	54.0	4.0	99300.0	80500.0	171000.0	70000.0

- Age:
  - Customers range from 22 to 54 years, with an average age of around 32 years.
  - Half of the customers fall between 25 and 38 years, indicating a largely young to mid career population.

- Number of Dependents:
  - Customers typically have 2 to 3 dependents, with an average of 2.46.
  - The narrow spread suggests fairly consistent family sizes across the dataset.

- Salary:
  - Individual salaries range from 30,000$ to 99,300$ with an average of about 60,400$.
  - Most customers earn between 51,900$ and 71,800$, showing a predominantly middle income customer base.

- Partner Salary:
  - Partner income varies widely, from 0$ to 80,500$, indicating that not all customers have a working partner.
  - The median partner salary of 25,100$ suggests that dual income households are common but not universal.

- Total salary:
  - Combined household income ranges from 30,000$ to 1,71,000, with an average close to 79,600$.
  - The interquartile range (60,500$ - 95,900$) highlights a strong concentration of customers in the mid income segment.

- Vehicle Price:
  - Vehicles price range from 18,000$ to 70,000$, with an average purchase price of approximately 35,600$.
  - Half of the customers spend between 25,000$ and 47,000$, indicating a preference for affordable to mid range vehicles.

5. Data Cleaning

5.1 NULL Values

5.1.1 Checking NULL values

	0
Age	0
Gender	53
Profession	0
Marital_status	0
Education	0
No_of_Dependents	0
Personal_loan	0
House_loan	0
Partner_working	0
Salary	0
Partner_salary	106
Total_salary	0
Price	0
Make	0

- The dataset is largely complete, with no missing values in most key variables.
- Gender has 53 missing entries.
- Partner_salary shows 106 missing values.

5.1.2 Treating Null Values

- Missing values in the Gender column were imputed using the most frequently occurring category, ensuring that the overall gender distribution of the dataset remains consistent.
- Missing values in the Partner_salary column were logically derived by subtracting the individual's salary from the total household salary, providing a realistic estimate where direct information was unavailable.

	0
Age	0
Gender	0
Profession	0
Marital_status	0
Education	0
No_of_Dependents	0
Personal_loan	0
House_loan	0
Partner_working	0
Salary	0
Partner_salary	0
Total_salary	0
Price	0
Make	0

5.2 Duplicate records

```
np.int64(0)
```

- A duplicate record check was performed across the dataset, and no duplicate entries were identified.
- This confirms that each record in the dataset is unique, ensuring the integrity of the data.

5.3 Logical inconsistencies

```
np.int64(0)
```

- A validation check was performed to ensure that Total Salary equals the sum of Salary and Partner Salary for each record.
- The results indicate that no logical inconsistencies were found across these salary related fields.

5.4 Invalid or unrealistic values

```
dtype: int64
```

- A check was conducted to identify any invalid or unrealistic values in the Salary, Partner Salary and Total Salary fields.
- The analysis confirmed that no negative values are present in any of these income related variables.

5.5 Consistency of Categorical Data

5.5.1. Checking Categorical Data

```
Unique values in Gender:
['Male' 'Femal' 'Female' 'Femle']

Unique values in Profession:
['Business' 'Salaried']

Unique values in Marital_status:
['Married' 'Single']

Unique values in Education:
['Post Graduate' 'Graduate']

Unique values in Personal_loan:
['No' 'Yes']

Unique values in House_loan:
['No' 'Yes']

Unique values in Partner_working:
['Yes' 'No']

Unique values in Make:
['SUV' 'Sedan' 'Hatchback']
```

- A consistency check was performed on all categorical variables by examining their unique values.
- Most categorical columns show clear and well defined categories, indicating consistent data entry.
- However, the Gender column contains inconsistent labels such as ‘Femal’ and ‘Femle’ in addition to ‘Male’ and ‘Female’.

5.5.2. Treating inconsistencies Categorical Data

```
array(['Male', 'Female'], dtype=object)
```

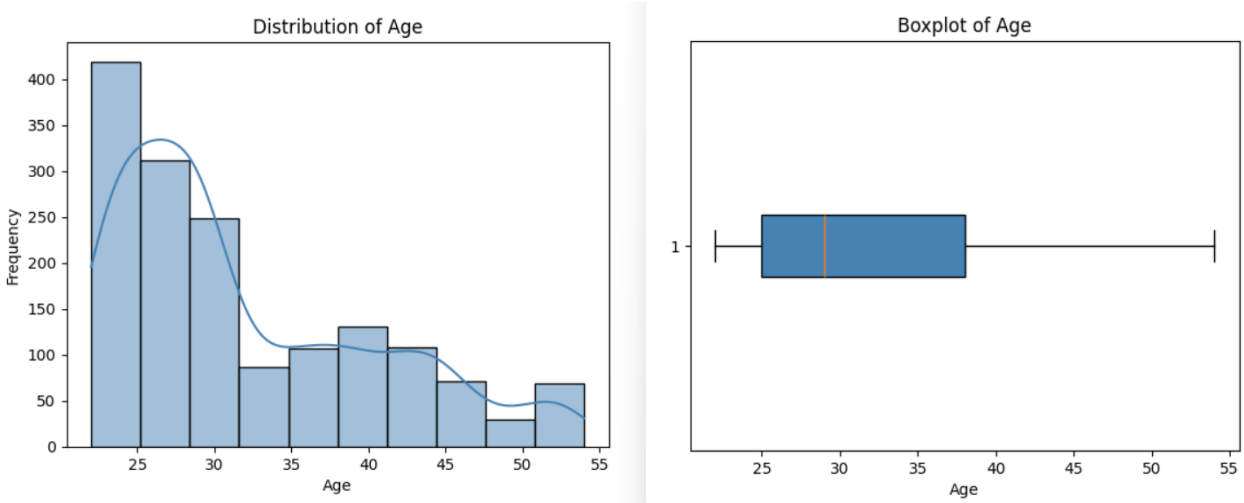
- The inconsistencies were corrected by standardizing all variations to the single category 'Female'.
- Post correction verification confirms that the Gender variable now contains only two valid and consistent categories: Male and Female
- This standardization improves data accuracy and ensures reliable gender based analysis in subsequent stages.

6. Exploratory Data Analysis

6.1. Univariate Analysis

6.1.1. Exploring all Numerical variables

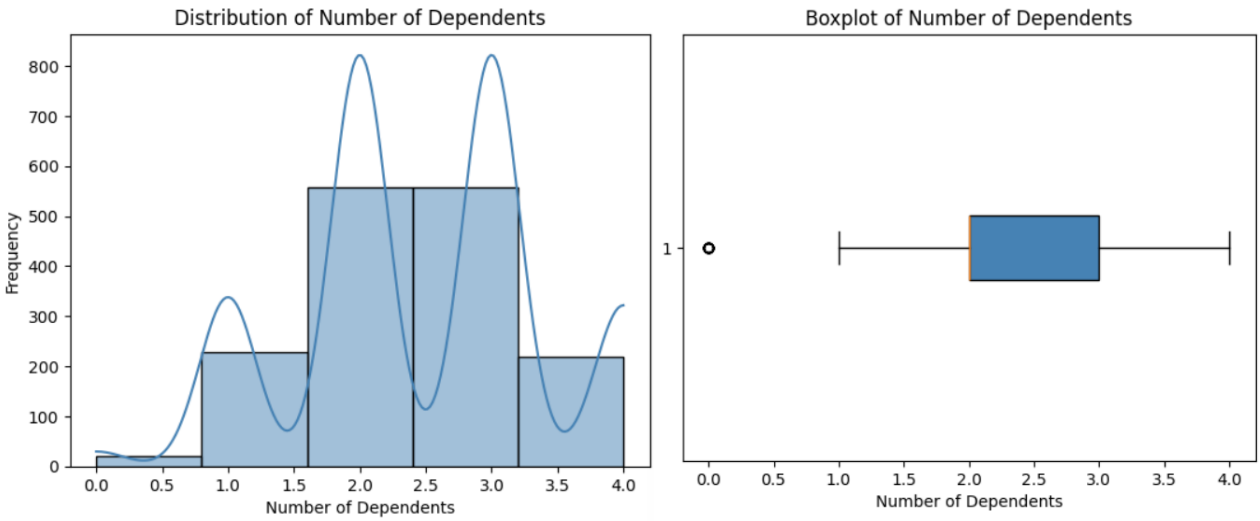
1. Age



Observation:

- The age distribution is concentrated among younger individuals, primarily between the mid-20s and mid-30s.
- The data shows a slight right skew, with fewer observations at higher age levels.
- Most customers fall within a moderate age range, indicating a predominantly working age population.
- The boxplot suggests limited extreme values, with no significant outliers impacting the distribution.

2. Number of Dependents

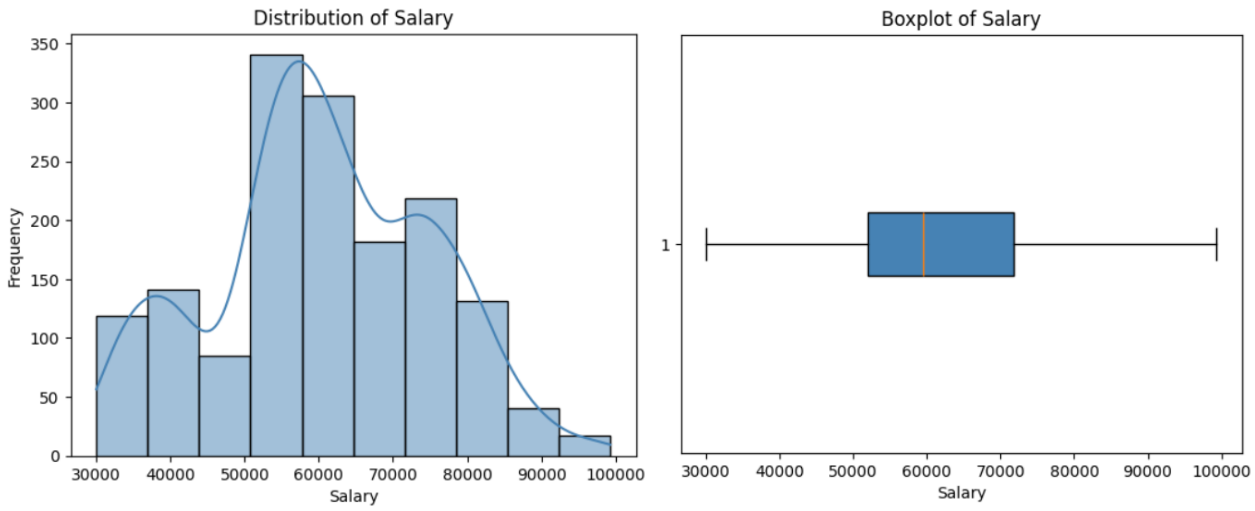


Observation:

- Most customers have 2 to 3 dependents, indicating moderate household sizes.
- Very few customers have no dependents, as seen from the lower frequency values.

- The distribution is fairly balanced with a slight concentration around the middle values.
- The boxplot shows limited variability with no significant extreme values affecting the data.

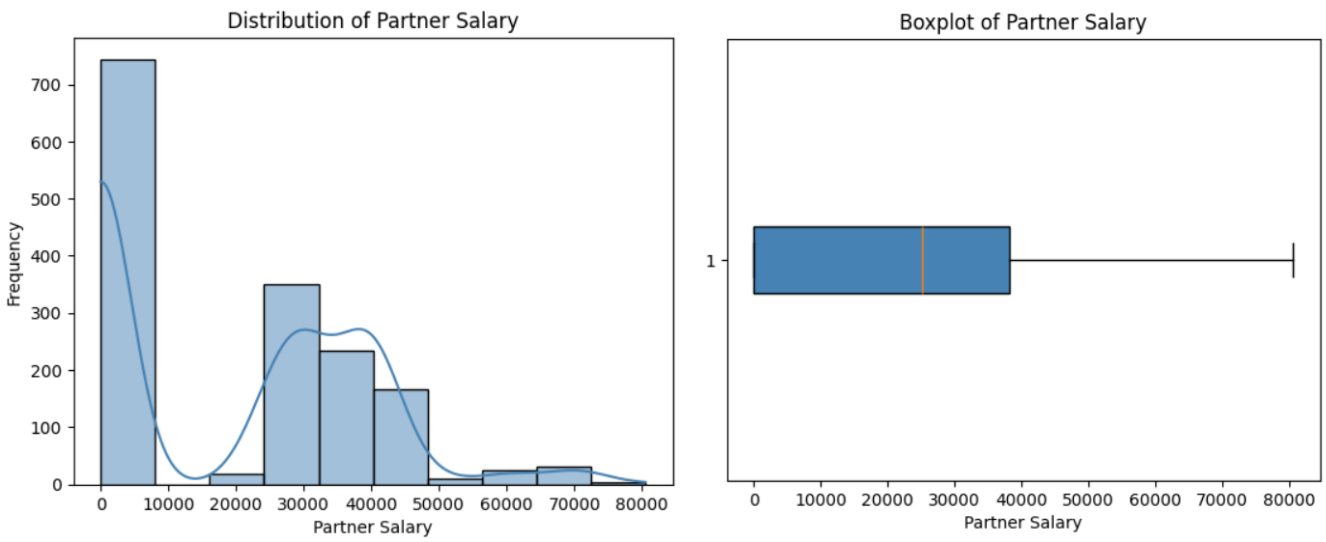
3. Salary



Observations:

- Salaries are mainly concentrated in the mid-income range, with most values between approximately 50,000$ and 75,000$.
- The distribution shows a slight right skew, indicating fewer high income individuals.
- Salary values display a moderate spread, reflecting income diversity among customers.
- The boxplot indicates no extreme outliers, suggesting stable and reliable salary data.

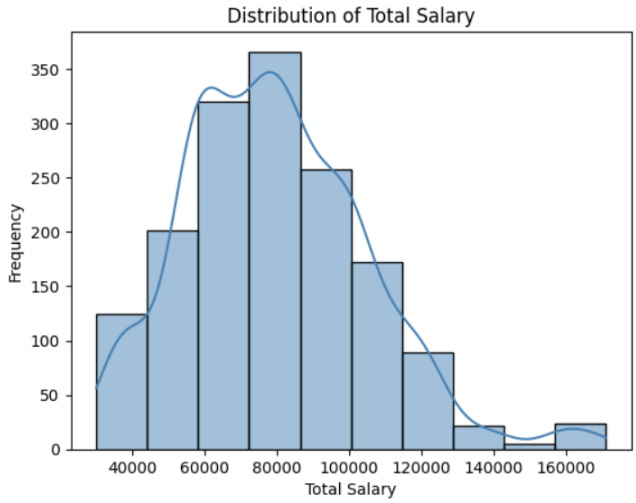
4. Partner Salary

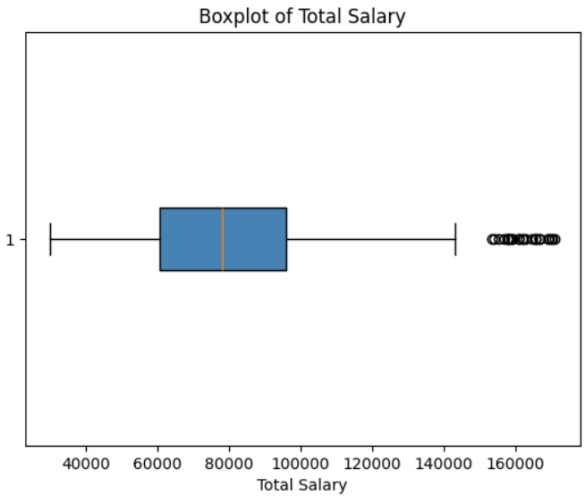


Observations:

- A large proportion of records show low or zero partner salary, indicating many single income households.
- Partner salary values display variability, ranging from zero to higher income levels.
- The distribution is right skewed, with fewer cases of high partner income.
- The boxplot indicates the presence of extreme values, though they represent a small portion of the data.

5. Total Salary

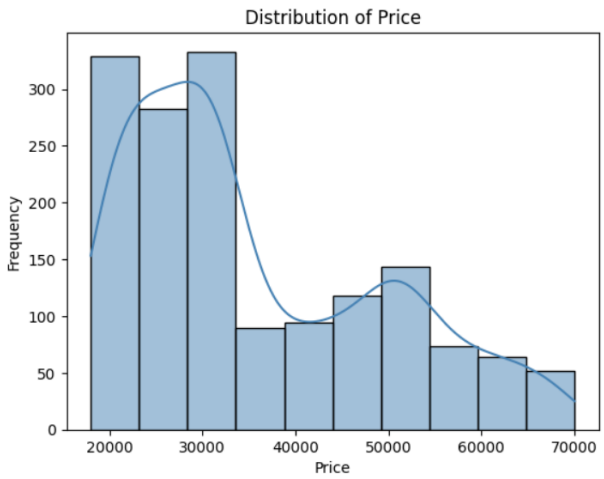


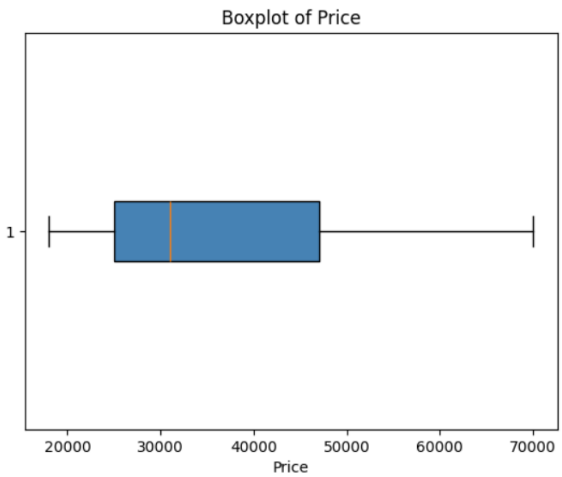


Observations:

- Total household income is mainly concentrated in the mid income range, indicating a majority of moderate earning households.
- The distribution shows a right skew, driven by a smaller group of high income households.
- Income levels exhibit considerable variation, reflecting diverse earning capacities across customers.
- The boxplot highlights the presence of high end outliers, representing a limited number of higher income cases.

6. Price





Observations:

- Vehicle prices are mainly concentrated in the lower to mid price range, indicating affordability focused customer segments.
- The distribution shows a right skew, with fewer high priced vehicles.
- Price levels exhibit a wide spread, reflecting multiple product offerings across proce points.
- The boxplot indicates some higher end values, though they represent a smaller portion of the dataset.

6.1.2. Exploring all Categorical variables

Frequency Distribution of Categorical Variables

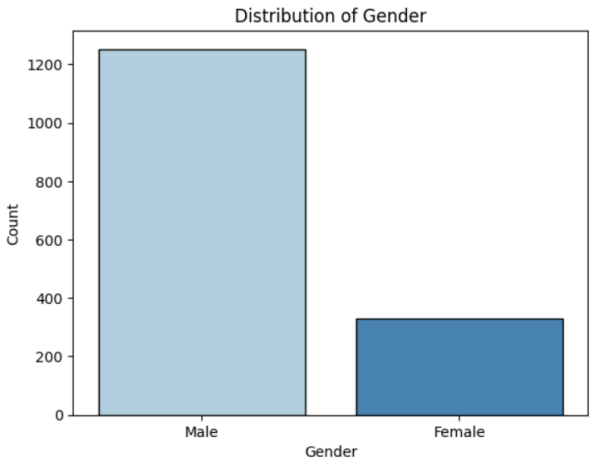
```
Value counts for Gender:
Gender
Male      1252
Female     329
Name: count, dtype: int64

Value counts for Profession:
Profession
Salaried    896
Business    685
Name: count, dtype: int64

Value counts for Marital_status:
Marital_status
Married    1443
Single      138
Name: count, dtype: int64

Value counts for Education:
Education
Post Graduate    985
Graduate         596
Name: count, dtype: int64
```

```
Value counts for Personal_loan:
Personal_loan
Yes    792
No     789
Name: count, dtype: int64

Value counts for House_loan:
House_loan
No     1054
Yes     527
Name: count, dtype: int64

Value counts for Partner_working:
Partner_working
Yes    868
No     713
Name: count, dtype: int64

Value counts for Make:
Make
Sedan        702
Hatchback    582
SUV          297
Name: count, dtype: int64
```

Observations:

- The customer base is predominantly male, with females representing a smaller segment.
- A larger proportion of customers are salaried professionals, followed closely by those in business.
- Most customers are married, indicating a family oriented customer profile.
- Post graduates form the majority, suggesting a relatively well educated customer base.
- Personal loan ownership is almost evenly split between customers with and without loans.
- A higher number of customers do not have a house loan, indicating lower long term debt exposure for most.
- In many households, the partner is working, contributing to overall household income.
- Sedans are the most preferred vehicle type, followed by hatchbacks, while SUVs form a smaller segment.

1. Gender

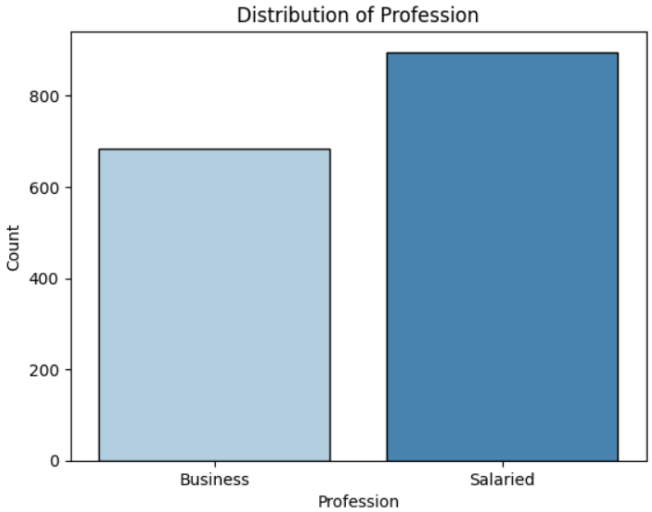


Observations:

- The customer base is predominantly male, with females forming a smaller segment.

- There is a noticeable gender imbalance, indicating higher representation of male customers.
- This distribution suggests potential opportunities for targeted outreach toward female customers.

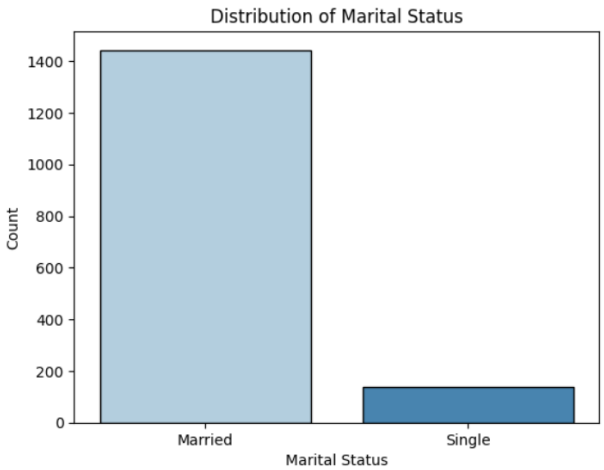
2. Profession



Observations:

- Salaried professionals form the larger share of the customer base.
- Business owners also represent a significant portion, though smaller than salaried customers.
- The distribution indicates a customer profile with stable and regular income sources, which is favourable for financial decision making.

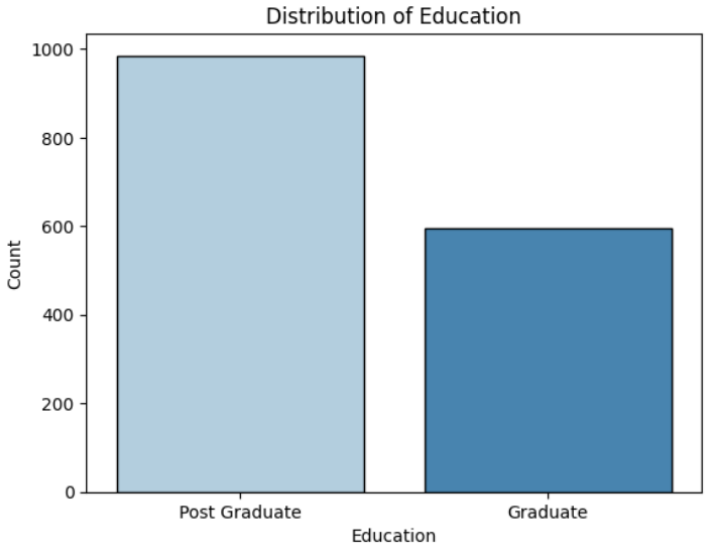
3. Marital Status



Observations:

- The majority of customers are married, forming the dominant segment.
- Single customers represent a much smaller proportion of the dataset.
- This suggests a large family oriented customer base, which may influence financial and purchasing decisions.

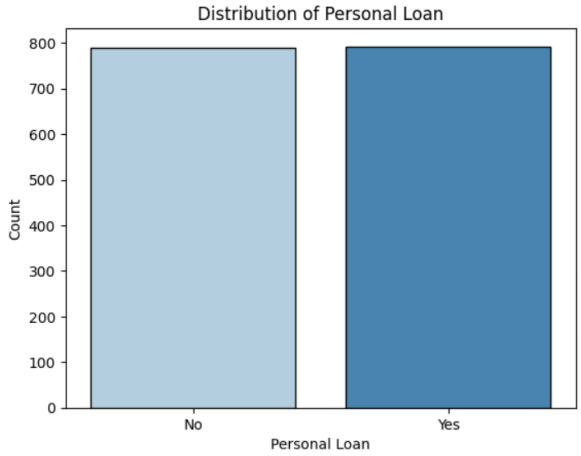
4. Education



Observations:

- Post graduates form the majority of the customer base.
- Graduates also represent a significant segment, though smaller in comparison.
- Overall, the dataset reflects a well educated customer profile, which may influence purchasing behaviour.

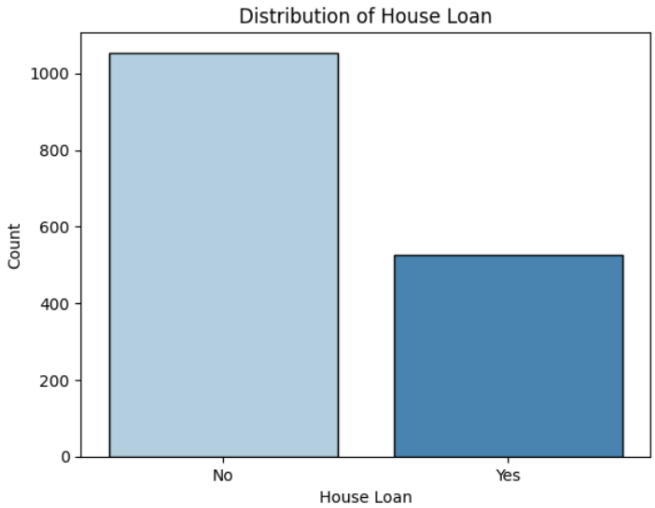
5. Personal Loan



Observations:

- Customers with and without personal loans are almost equally represented.
- This indicates a balanced mix of borrowing and non borrowing customers.
- The distribution suggests potential opportunities for targetted loan offerings across both segments.

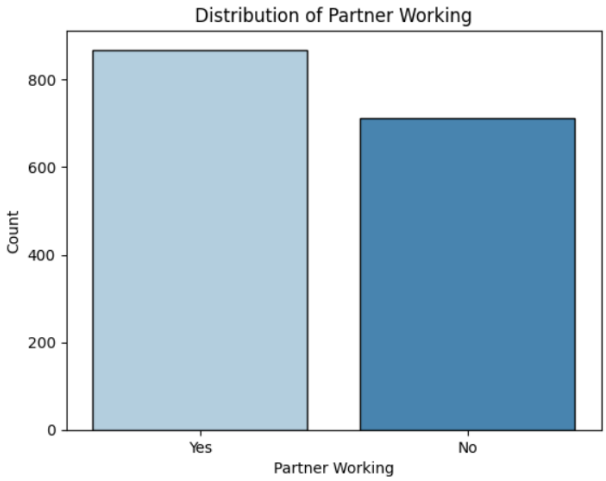
6. House Loan



Observations:

- A majority of customers do not have a house loan.
- A smaller but notable segment of customers have an existing house loan.
- This suggests relatively lower long term housing debt among most customers.

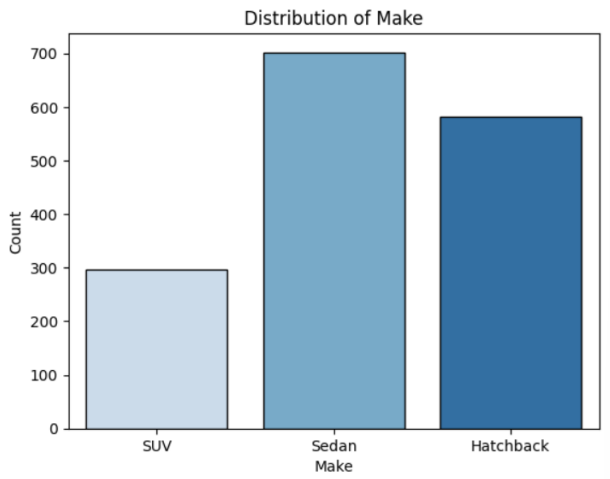
7. Partner Working



Observations:

- A slight majority of customers have a working power.
- A significant portion of customers have a non working partner, indicating single income households.
- This distribution highlights variation in household income structures.

8. Make



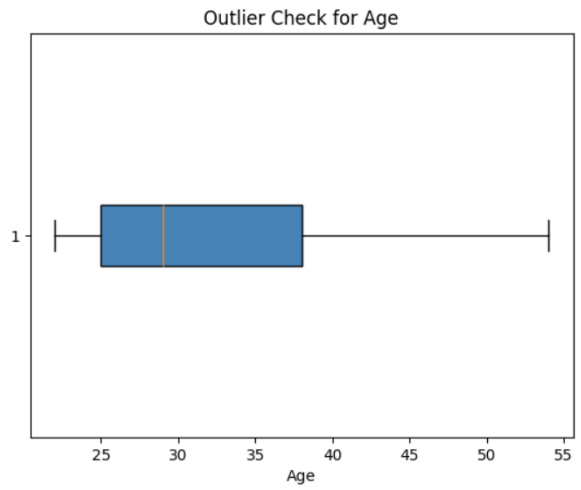
Observations:

- Sedans are the most preferred vehicle type among customers.
- Hatchbacks also have strong representation, indicating demand for compact vehicles.
- SUVs form the smallest segment, suggesting comparatively lower preferences for larger vehicles.

6.1.3. Outliers Detection & Treating them

1. Visual Inspection of Outliers using Boxplot

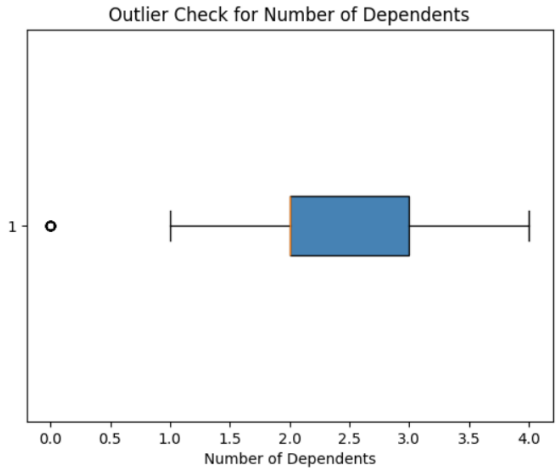
A) AGE



Observation:

- The age values are mostly concentrated within a reasonable range.
- No significant extreme outliers are observed in the age distribution.
- The data appears stable and suitable for further analysis without outlier treatment.

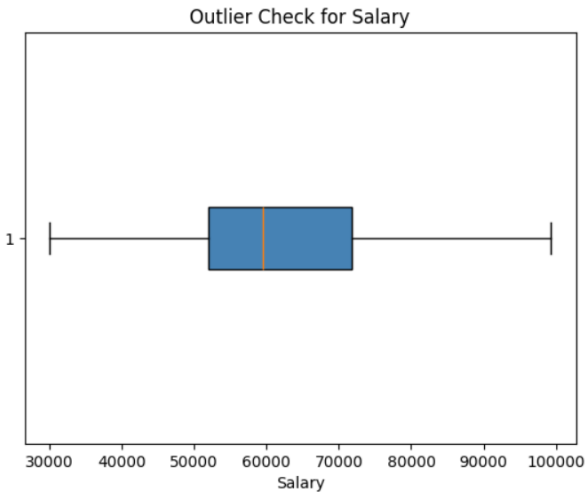
B) NUMBER OF DEPENDENTS



Observations:

- Most values fall within the expected range of 1 to 4 dependents.
- A small number of low end outliers(zero dependents) are observed.

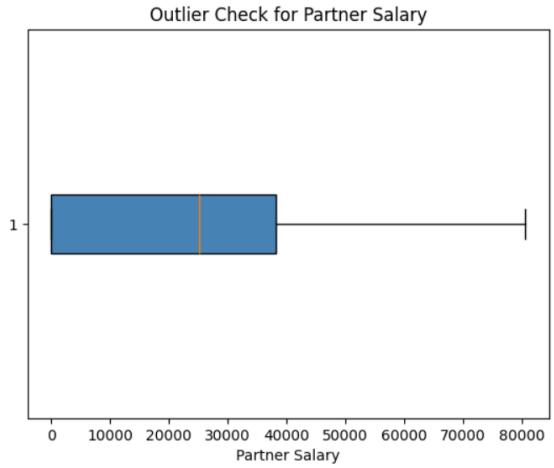
C) SALARY



Observations:

- Salary values are largely within a consistent and reasonable range.
- No significant extreme outliers are observed in the salary distribution.

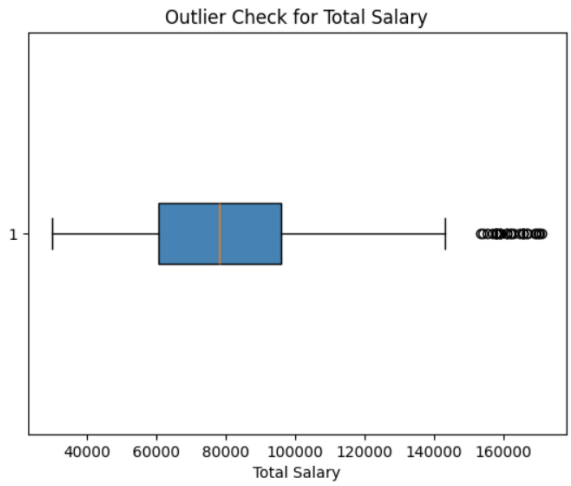
D) PARTNER SALARY



Observations:

- Partner salary values show a wide spread, including zero income cases.
- A few high value outliers are visible are visible on the upper end of the distribution.

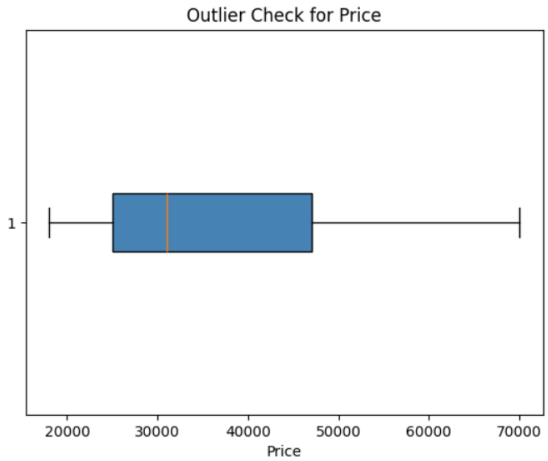
E) TOTAL SALARY



Observations:

- Most total salary values lie within a reasonable and consistent range.
- A small number of high income outliers are present on the upper end of the distribution.
- These outliers represent high earning households.

F) PRICE



Observations:

- Price values are distributed within a reasonable and expected range.
- No extreme outliers are observed in vehicle prices.

2. Detecting outliers using IQR method

- Outliers were identified using the Interquartile Range (IQR) method, which is a robust technique commonly used for numerical data.
- For each numerical variable, the first quartile (Q1) and third quartile (Q3) were calculated.
- The IQR was computed as the difference between Q3 and Q1 (IQR + 1.5 * IQR) was classified as an outlier.

```
Age                 0
No_of_Dependents   20
Salary              0
Partner_salary      0
Total_salary       27
Price               0
dtype: int64
```

Observations:

- No outliers were detected in Age, Salary, Partner_salary, or Price using the IQR method.
- Number of Dependents shows the presence of 20 outliers, indicating a few uncommon household sizes.
- Total_salary contains 27 outliers, suggesting some customers have significantly higher combined incomes compared to the rest of the dataset.
- Overall, outliers are limited to specific variables and do not affect the majority of numerical features.

3. Treating Outliers

- Outliers in the Number of Dependents column & Total Salary column were handled using the Interquartile Range (IQR) capping method.
- The first (Q1) and third (Q3) quartiles were calculated to understand the spread of typical values.
- Based on these, lower and upper bounds were defined as Q1-1.5*IQR and Q3+1.5*IQR, respectively.

- Any values falling outside this range were capped to the nearest boundary instead of being removed.
- This approach ensured that extreme values did not disproportionately influence the analysis while preserving all records in the dataset.

6.1.3. Key Observations

- The dataset shows a balanced age distribution, with most customers falling within the young to middle-aged working population, indicating a strong representation of active car buyers.
- Gender distribution is skewed towards male customers, suggesting that men form a larger share of the current customer base.
- In terms of profession, salaried individuals dominate the dataset, highlighting them as the primary target segment for automobile purchases.
- A significant majority of customers are married, which aligns with the presence of family oriented purchase behavior.
- Most customers are either graduates or post graduates, indicating a relatively educated customer base.
- The number of dependents is concentrated around 2 to 3, reflecting typical household sizes.
- Personal and housing loan statuses show a mix of borrowers and non-borrowers, suggesting varies financial commitments among customers.
- Salaries and total household income exhibit a wide range, capturing both budget conscious and higher spending customers.
- Vehicle prices also span a broad range, indicating demand cross entry level to premium segments.
- Among vehicle types, Sedans and Hatchbacks are more commonly purchased compared to SUVs, pointing toward higher demand for practical and cost effective models.

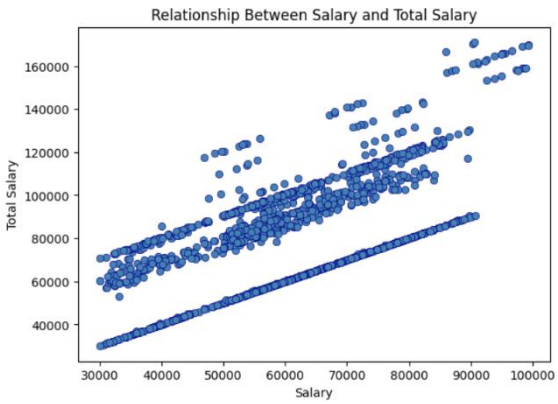
6.2. Bivariate Analysis

6.2.1. Exploring the relationship between all numerical variables.

1. Salary vs Total Salary

Observations:

- We can see as individual salary increases, total household income also increases, showing a clear and expected trend.
- The upward pattern in the plot indicated that salary in a key driver of total income.
- The spread of points suggests that partner income adds to the variation in total salary.

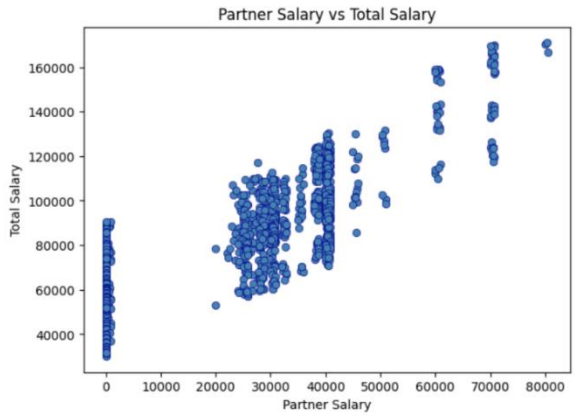


2. Partner Salary vs Total Salary

Observations:

- Total household income generally increases as partner salary increases, showing a clear upward trend.

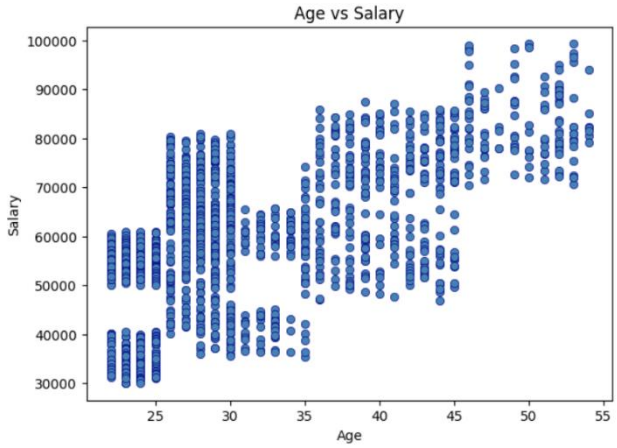
- Households with no partner income cluster at lower total salary levels, forming a distinct group.
- The spread of points suggests that partner income significantly adds to overall household earnings.



3. Age vs Salary

Observations:

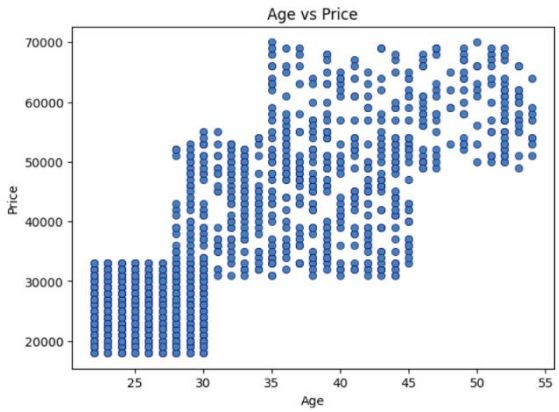
- Salary generally tends to increase with age, reflecting career progression over time.
- Younger individuals mostly fall into lower to mid salary ranges, while older customers show higher earning levels.
- There is noticeable spread at most ages, indicating that income varies even within the same age group.
- Overall, the relationship appears positive but not perfectly linear.



4. Age vs Price

Observations:

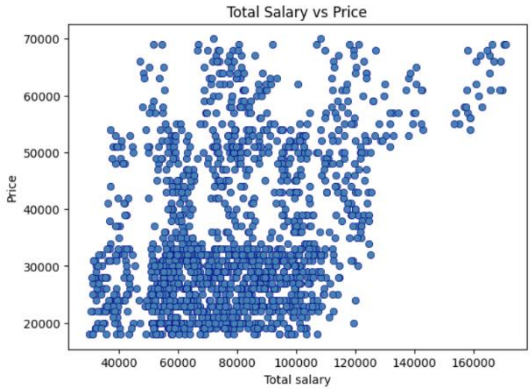
- Vehicle price generally increases with age, showing an upward trend.
- Younger customers are mostly clustered around lower priced vehicles.
- As age increases, the price range becomes wider, indicating varied purchasing power.
- Overall, the relationship appears positive but not perfectly linear.



5. Salary vs Price

Observations:

- Customers with higher total household income tend to consider higher priced vehicles, though the trend is not very tight.
- A large number of customers cluster in the lower to mid price range, regardless of income level.
- The spread of points suggests that factors beyond income also influence vehicle price decisions.

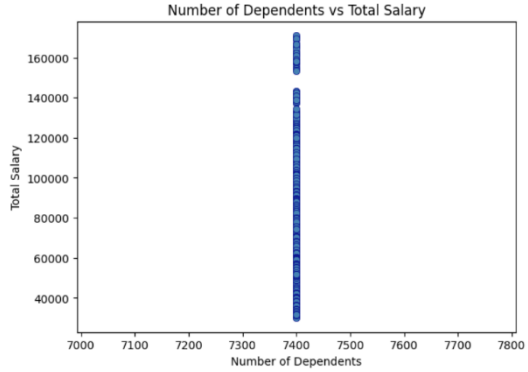


- Overall, the relationship appears positive but moderate, rather than strongly linear.

6. Number of Dependents vs Total Salary

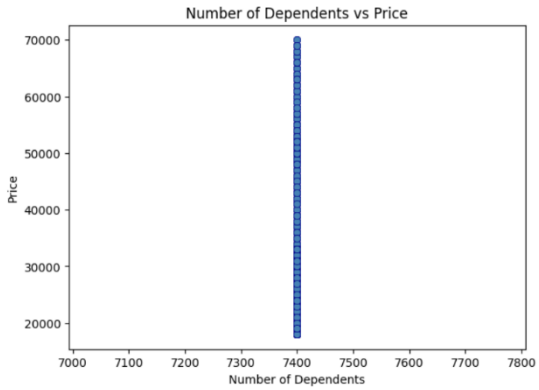
Observations:

- We cannot see any clear pattern between the number of dependents and total household income.
- Income values are widely spread across the same dependent count, indicating high variability.
- This suggests that the number of dependents does not directly influence total salary in the dataset.



7. Number of Dependents vs Price

- Vehicle prices are spread across all levels of dependents, with no visible trend.
- Customers with the same number of dependents purchase vehicles across a wide price range.
- This indicates that family size does not strongly influence vehicle price decisions.



6.2.2. Exploring the correlation between all numerical variables.

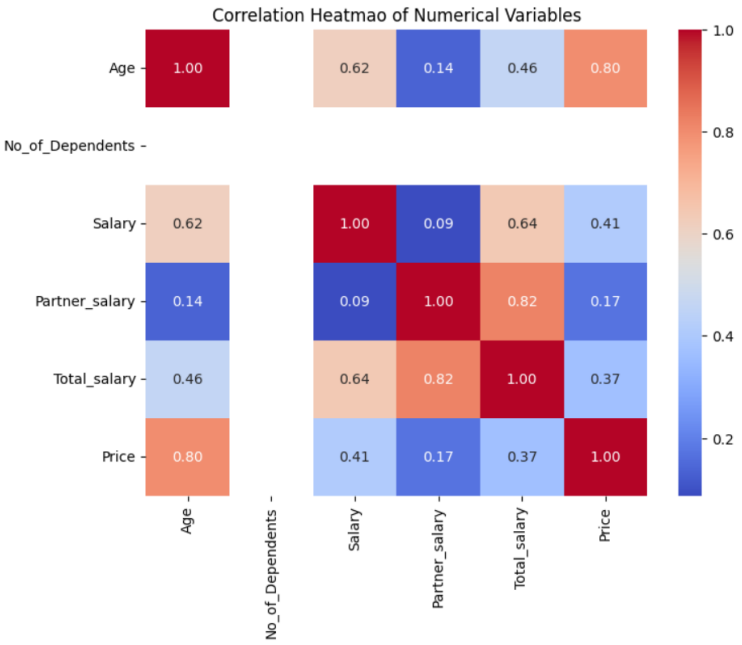
1. Correlation matrix for numerical variables

	Age	No_of_Dependents	Salary	Partner_salary	Total_salary	Price
Age	1.000000	NaN	0.616899	0.135702	0.458869	0.797831
No_of_Dependents	NaN	NaN	NaN	NaN	NaN	NaN
Salary	0.616899	NaN	1.000000	0.087155	0.641560	0.409920
Partner_salary	0.135702	NaN	0.087155	1.000000	0.820069	0.171875
Total_salary	0.458869	NaN	0.641560	0.820069	1.000000	0.367823
Price	0.797831	NaN	0.409920	0.171875	0.367823	1.000000

Observations:

- Salary and Total Salary show a strong positive correlation, which is expected since total salary is largely driven by the individual's income.
- Partner Salary has a very strong relationship with Total Salary, indicating that households with earning partners significantly increase overall purchasing power.
- Age is moderately correlated with Salary, suggesting that income generally increases with age, likely due to greater work experience.
- Age also shows a strong positive correlation with Vehicle Price, implying that older customers tend to purchase higher priced vehicles.
- Vehicle Price has a moderate correlation with both Salary and Total Salary, indicating that higher income customers are more likely to spend more on automobiles.
- Number of Dependents shows no meaningful correlation with income or vehicle price, suggesting it does not strongly influence purchasing capacity in this dataset.

2. Correlation Heatmap



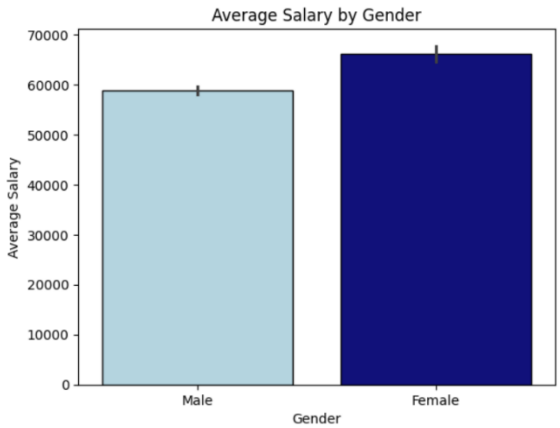
Observations:

- Age shows a strong positive correlation with vehicle price (0.80), indicating that older customers tend to purchase higher priced vehicles.
- Salary and Total Salary are moderately to strongly correlated (0.64), which is expected since total income includes the individual's salary.
- Partner Salary has a very strong correlation with Total Salary (0.82), highlighting the significant role of dual incomes in increasing overall purchasing power.
- Age has a moderate correlation with Salary (0.62), suggesting income generally increases with age up to a certain point.
- Vehicle Price has a moderate relationship with Salary (0.41) and Total Salary (0.37), implying income influences car choice, but is not the only deciding factor.
- Number of dependents shows no meaningful correlation with income or price, indicating family size does not strongly affect vehicle pricing decisions.

6.2.3. Exploring the relationship between Categorical & Numerical variables.

1. Gender vs Numerical Variables

A) GENDER VS SALARY



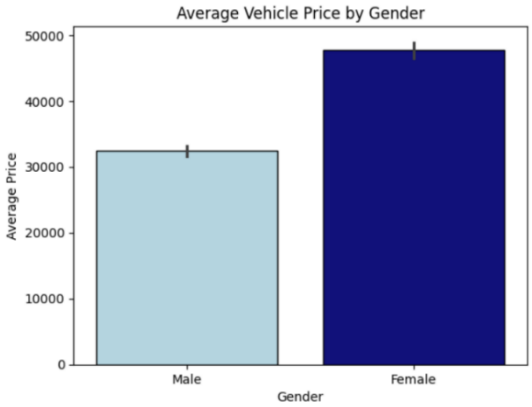
Observations:

- Female customers show a slightly higher average salary compared to male customers.
- The difference in average salary between genders is present but not very large.

B) GENDER VS PRICE

Observations

- Female customers tend to purchase higher priced vehicles on average compared to male customers.
- The difference in average vehicle price between genders is quite noticeable.
- This suggests that gender based preferences or purchasing capacity may influence vehicle selection in this dataset.



2. Profession vs Income

A) PROFESSION VS SALARY

- Salaried individuals show a slightly higher average salary compared to those in business.
- The difference in average salary between the two professions is small, indicating comparable earning levels.
- Overall, profession does not appear to create a major salary gap in this dataset.

B) PROFESSION VS TOTAL SALARY

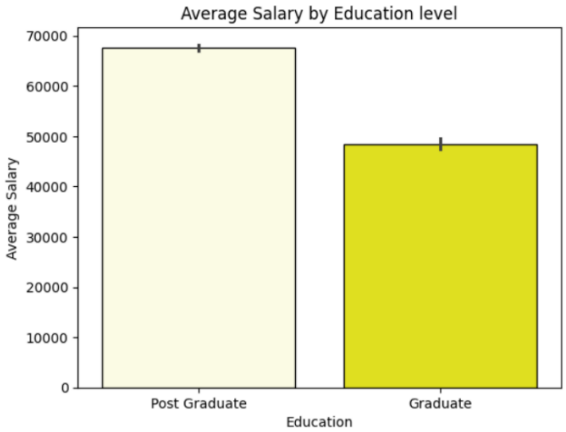
- Salaried customers have a slightly higher average total household income compared to business customers.
- The difference in total salary between the two professions is not very large, indicating similar overall earning capacity.

3. Education vs Numerical Variables

A) EDUCATION VS SALARY

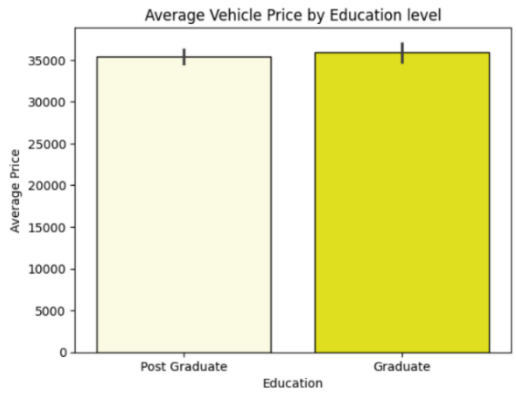
Observations:

- Post graduates customers have a noticeably higher average salary compared to graduates.
- The difference suggests that higher education is associated with better earning potential in this dataset.
- Overall, education level appears to have a clear impact on salary levels.



B) EDUCATION VS PRICE

- The average vehicle price is very similar across both education levels.
- There is no significant difference in vehicle spending between graduates and post graduates.
- This suggests that education level alone does not strongly influence vehicle price decisions in this dataset.

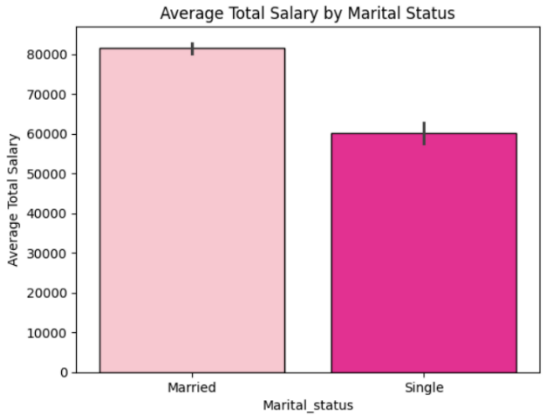


4. Marital Status vs Numerical Variable

A) MARITAL STATUS VS TOTAL SALARY

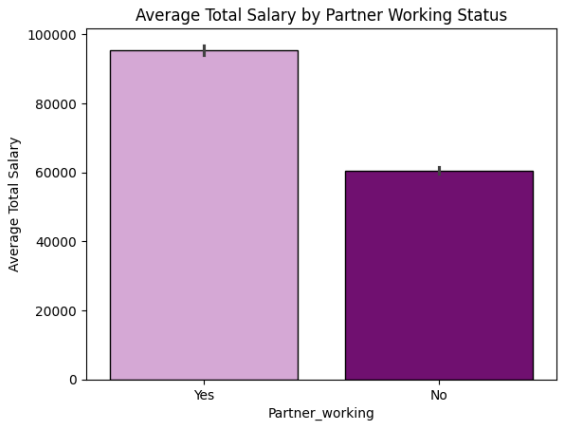
Observations:

- Married customers have a higher average total household income compared to single customers.
- This difference likely reflects the presence of dual income households among married individuals.
- Overall, marital status appears to have a noticeable impact on total salary in this dataset.



5. Partner Working vs Income

- Households where the partner is working have a significantly higher average total income.
- The difference clearly highlights the impact of dual income households on overall earnings.
- Partner working status appears to be a strong factor influencing total salary in this dataset.

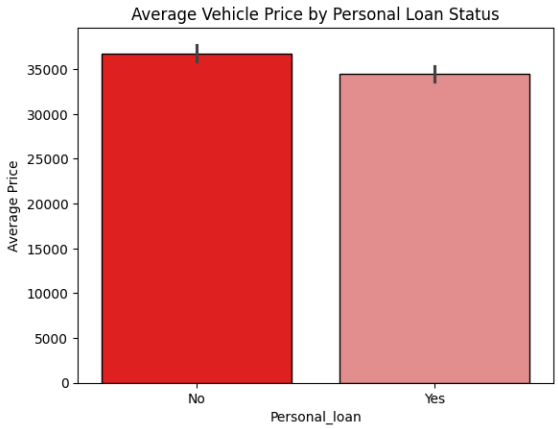


6. Loan Status vs Price

A) PERSONAL LOAN VS PRICE

Observations:

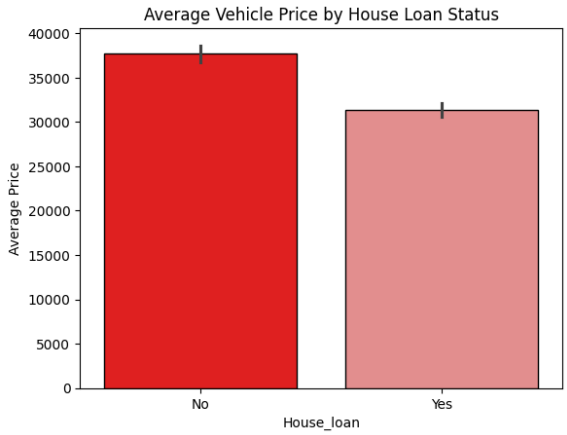
- Customers who did not take a personal loan tend to purchase slightly higher priced vehicles compared to those who took a loan.
- This suggests that loan takers may be more price conscious when making automobile purchases.



B) HOUSE LOAN VS PRICE

Observations:

- Customers without a personal loan purchase slightly higher priced vehicles on average.
- The difference in average vehicle price between loan holders and non loan holders is very large.
- This suggests that personal loan status does not strongly influence vehicle price decisions in this dataset.

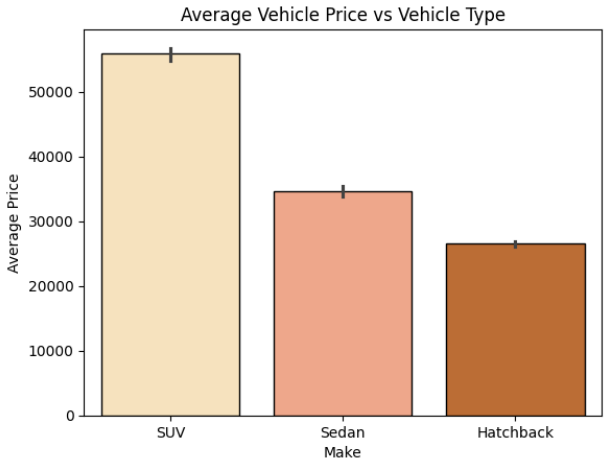


7. Vehicle Make vs Numerical Variables

A) MAKE VS PRICE

Observations:

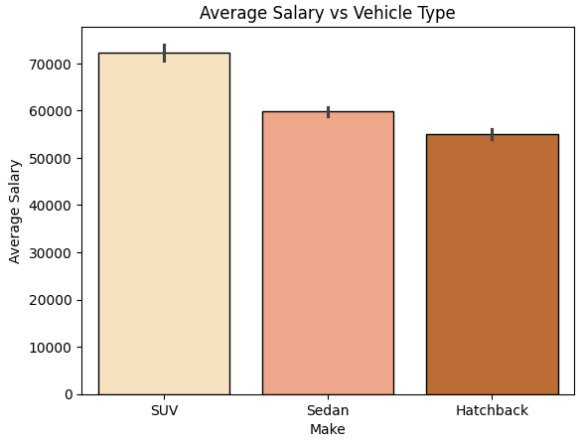
- SUVs have the highest average price, clearly positioned as a premium vehicle segment.
- Sedans fall in the mid price range, balancing affordability and features.
- Hatchbacks are the most affordable option, targeting budget-conscious customers.
- The clear price gap between vehicle types shows well defined market segmentation, which can help in designing focused marketing strategies.



B) MAKE VS TOTAL SALARY

Observations:

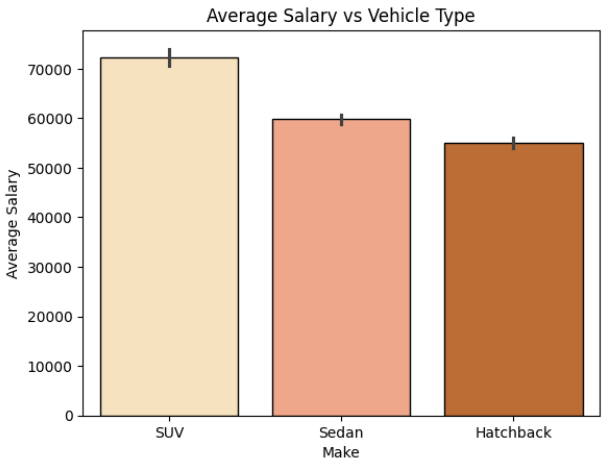
- Customers choosing SUVs have the highest average total salary, indicating higher purchasing power.
- Sedan buyers fall in the middle, suggesting a balance between affordability and income level.
- Hatchback buyers have the lowest average total salary, pointing toward a more budget conscious segment.
- This pattern shows that vehicle choice strongly aligns with customer income, which is useful for targeted marketing and pricing strategies.



c) MAKE VS SALARY

Observations:

- SUV buyers have the highest average salary, suggesting they belong to a higher income group.
- Sedan buyers fall in the middle, indicating moderate earning capacity.
- Hatchback buyers have the lowest average salary, which shows that cost plays a bigger role in their purchase decision.



6.2.4. Key Insights

1. Relationship between Numerical Variables

- **Salary vs Total Salary:** A strong positive relationship is observed, indicating that an increase in individual salary directly contributes to a higher total household income. This is expected, as total salary is derived from individual and partner earnings.
- **Partner Salary vs Total Salary**: Partner Salary shows a very strong association with total salary, highlighting that households with dual incomes generally have significantly higher purchasing power.
- **Age vs Salary**: Salary tends to increase with age, especially from early career stages to mid career, suggesting income growth with experience and job stability.
- **Age vs Vehicle Price**: Older customers generally purchase higher priced vehicles, indicating a shift toward premium or higher segment cars as financial stability increases.
- **Total Salary vs Vehicle Price**: A moderate positive relationship exists, suggesting that customers with higher household income are more likely to spend more on vehicles.
- **Number of Dependents vs Income:** No strong or consistent relationship is observed. Vehicle price and total salary vary widely across different dependent counts, indicating that dependents alone do not significantly influence purchasing capacity.

2. Gender-Based Comparisons

- **Gender vs Salary**: Female customers show a slightly higher average salary compared to male customers in this dataset.
- **Gender vs Vehicle Price**: Female customers, on average, spend more on vehicle purchases than male customers, suggesting differences in purchasing preferences or financial decision making.

3. Profession Based Insights

- **Profession vs Salary**: Salaried individuals earn slightly more on average than business owners, indicating more stable income patterns among salaried customers.
- **Profession vs Total Salary**: Salaried households also show marginally higher total household income, making them an important target segment for premium offerings.

4. Education Level vs Purchasing Behaviour

- **Education vs Salary**: Post graduates earn significantly higher average salaries compared to graduates, reflecting the impact of higher education on income levels.

- **Education vs Vehicle Price**: Customers with higher education levels tend to purchase slightly more expensive vehicles, indicating a correlation between education, income and spending behaviour.

5. Marital Status vs Partner Influence

- **Marital Status vs Total Salary**: Married individuals have substantially higher total household income, largely due to dual incomes.
- **Partner Working vs Total Salary**: Customers with a working partner exhibit much higher total salaries, which directly influences their ability to purchase higher priced vehicles.

6. Loan Status and Spending Behaviour

- **Personal Loan vs Vehicle Price**: Customers who have not taken a personal loan spend more on vehicles compared to those who have taken loans, suggesting greater financial comfort.
- **House Loan vs Vehicle Purchase**: Customers with existing house loans tend to purchase relatively lower priced vehicles, possibly due to ongoing financial commitments.

7. Vehicle Type and Financial Capacity

- **Vehicle Type vs Salary**: SUV buyers have the highest average salary, followed by Sedan and Hatchback buyers.
- **Vehicle Type vs Total Salary**: SUV buyers also show the highest total household income, indicating that SUVs appeal more to financially stronger households.
- **Vehicle Type vs Price**: SUVs are the most expensive category, while Hatchbacks are the most affordable, clearly segmenting customers based on income levels.

6.2.5 Overall Interpretation

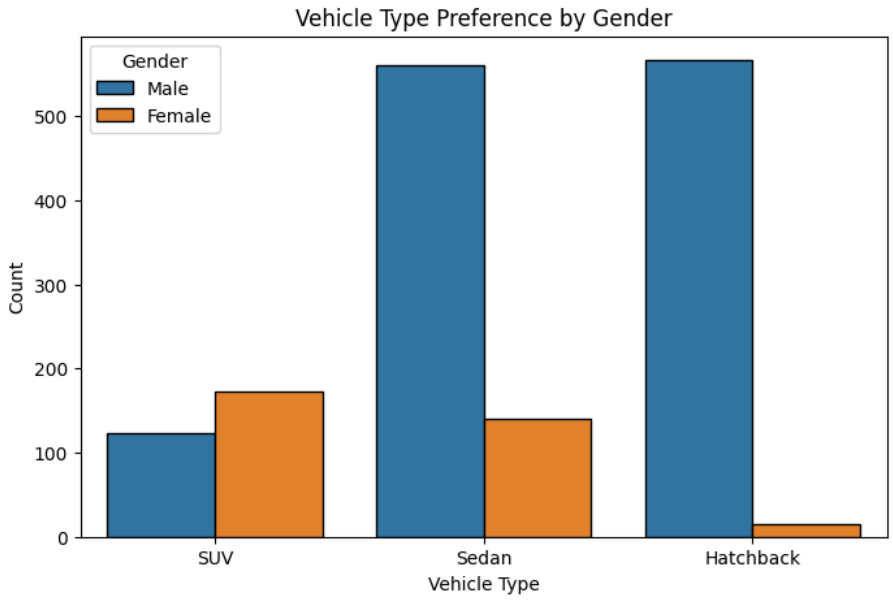
The bivariate analysis reveals that income related variables (salary, partner salary, total salary) play a crucial role in determining vehicle type and price preferences. Demographic factors such as age, education, marital status, and partner employment significantly influence purchasing power, while loan status reflects financial sensitivity.

The insights will help the Austo Motor Company in the following ways:

- Target SUV campaigns toward high income, married, dual income households.
- Promote Sedans to salaried professionals with moderate income.
- Position Hatchbacks for younger, budget conscious customers.

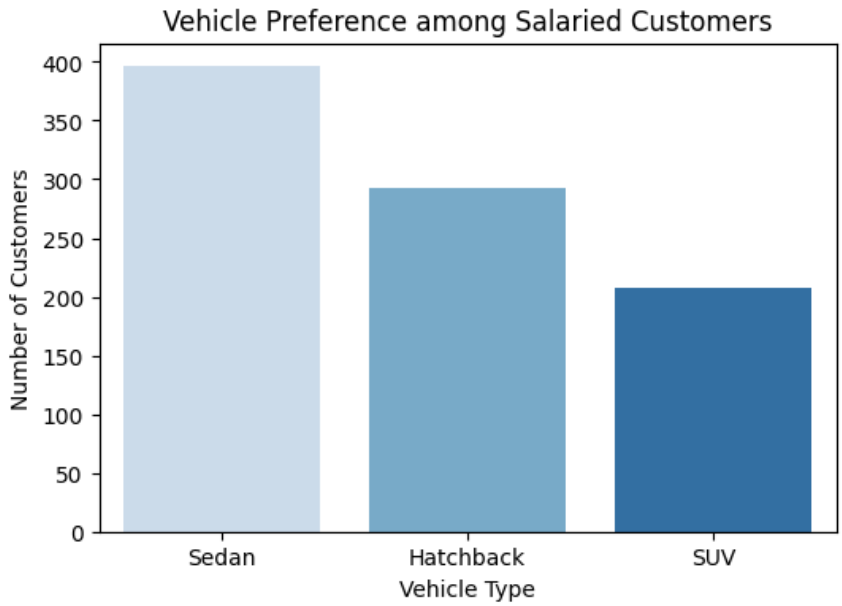
7. Key Questions

7.1. Do men tend to prefer SUVs more compared to women?



Based on the chart, men do not tend to prefer SUVs more than women. The number of female customers choosing SUVs is visibly higher than that of male customers. In contrast, men show a much stronger preference for Sedans and Hatchbacks, where their counts are significantly higher. This suggest that SUV demand in this dataset is relatively stronger among women, while men lean more toward non-SUV vehicle types.

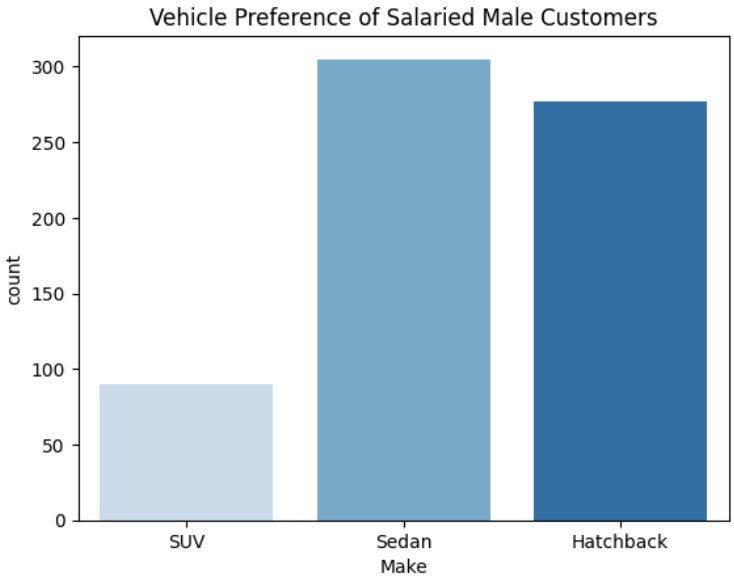
7.2. What is the likelihood of a salaried person buying a Sedan?



From the graph, its clear that Sedans are the most preferred choice among salaried customers. The Sedan bar is noticeably higher than both Hatchback and SUV, indicating that a large proportion of salaried individuals

are likely to buy a Sedan. This suggests that salaried customers tend to favour Sedans, likely because they strike a good balance between price, comfort and practically for regular use.

7.3. What evidence or data supports Sheldon Cooper's claim that a salaried male is an easier target for a SUV sale over a Sedan sale?

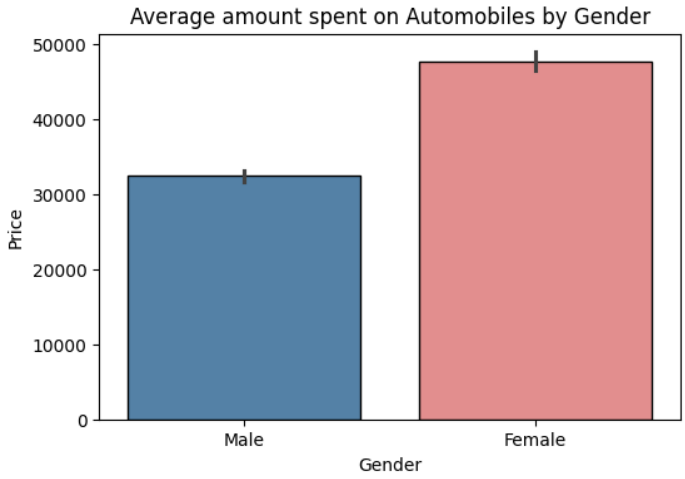


Based on the graph, there is no evidence to support Sheldon Cooper's claim.

The chart clearly shows that salaried male customers prefer Sedans far more than SUVs. The Sedan bar is the highest, while SUVs have the lowest count among the three vehicle types. This indicates that salaried males are more inclined toward Sedans rather than SUVs, making them a less likely target for SUV sales compared to Sedan sales.

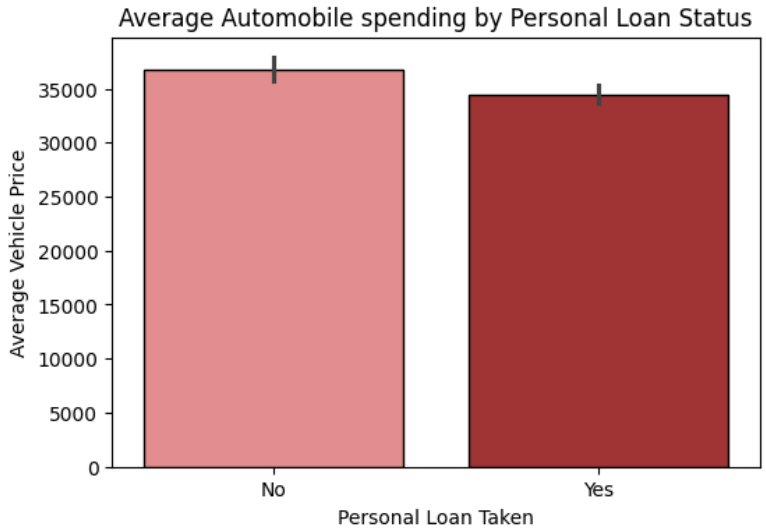
Hence, the data contradicts the claim and suggests that Sedans are the easier sale for salaried male customers, not SUVs.

7.4. How does the amount spent on purchasing automobiles vary by gender?



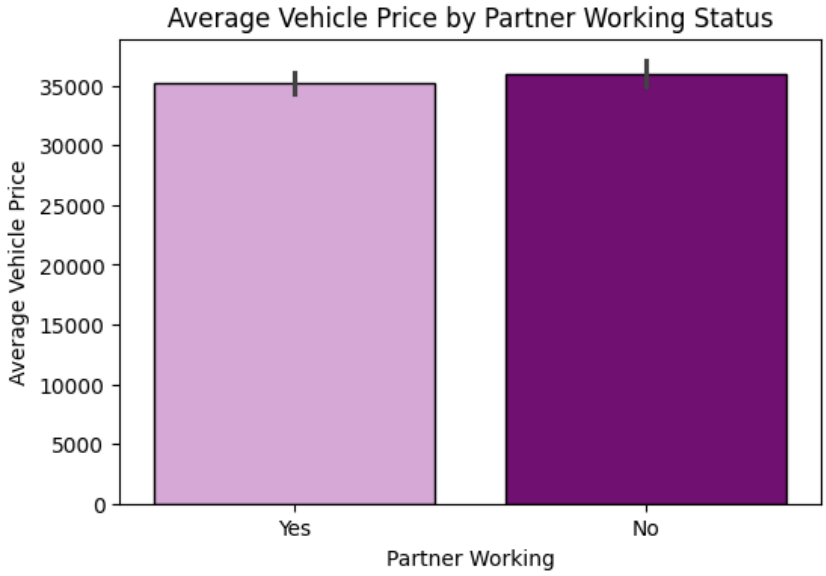
From the graph, it is clear that female customers spend more on automobiles on average than male customers. The average vehicle price for females is noticeably higher, indicating a tendency among women to opt for higher priced vehicles compared to them. This suggests that gender plays a role in spending behaviour, with female buyers showing a higher average purchase value in the automobile segment.

7.5. How much money was spent on purchasing automobiles by individuals who took a personal loan?



Based on the graph that we got, individuals who took a personal loan spent slightly less on automobiles on average. Their average spending is around 34,000$ - 35,000$, which is marginally lower compared to those who did not take a personal loan, indicating a more cautious or budget conscious purchase behaviour among loan takers.

7.6. How does having a working partner influence the purchase of higher-priced cars?



From the graph, having a working partner does not significantly change the amount spent on purchasing a car. Customers with a working partner and those without one show very similar average vehicle prices, with only a slight increase for those whose partner is not working. This suggests that partner employment status has limited influence on the decision to buy higher priced cars.

8. Actionable Insights

A close look at the data shows that Austo Motor's customer base is largely made up of young to middle aged working professionals, with most customers falling between their mid-20s and late-30s. The majority of buyers earn moderate income levels, have two to three dependents, and belong to salaried professions. Sedans emerge as the most commonly purchased vehicle type, followed by Hatchbacks, while SUVs represent a smaller but distinct segment. This overall distribution highlights that affordability, practicality and day-to-day usability are key drivers of demand for most customers.

When we examine relationships between variables, clearer patterns begin to emerge. Income, especially total household income, shows a strong link with vehicle price, indicating that customers with higher combined earnings are more willing and able to spend on costlier cars. Age also plays an important role, with older customers generally opting for higher priced vehicles, suggesting increased financial stability and evolving lifestyle needs over time. Gender based analysis reveals that female customers, on average, spend more on automobiles than male customers, pointing towards a stronger preference for premium features or higher value models.

Vehicle preference further varies across customer segments. Salaried individuals tend to favour Sedans, reinforcing the idea that comfort and value are prioritized in this group. Hatchbacks attract buyers with relatively lower income levels, making them a more price sensitive segment. SUVs, while less frequently purchased, are associated with customers who have higher total income and are willing to invest more, indicating that SUVs function more as an aspirational purchase rather than a mass market choice.

Financial commitments also influence buying behaviour. Customers without personal loans generally spend slightly more on vehicles, suggesting greater financial flexibility. Meanwhile, households with a non working partner tend to spend marginally more on vehicles, possibly reflecting single high income earners making consolidated household purchase decisions.

8.1. Brief Summary on Insights.

To conclude, both univariate and bivariate insights suggest that Austo Motor's marketing strategy should move away from generic messaging and toward customer specific targeting. Sedans and Hatchbacks should be positioned as practical, value driven options for salaried and mid income buyers, while SUVs should be marketed as premium, lifestyle oriented products aimed at older customers and higher income households. Gender specific campaigns highlighting premium features may also resonate better with female buyers. By aligning vehicle positioning with income, age, and household dynamics, the company can improve campaign effectiveness, customer satisfaction, and overall conversion rates.

9. Business Recommendations

Based on the customer patterns observed, Austo Motor Company should move toward a more targeted and customer centric marketing approach rather than relying on broad, generic campaigns.

- Segment marketing by income and life stage:
  Customers with higher household income and older age groups tend to spend more on vehicles and are more inclined toward premium models. Marketing campaigns for SUVs should focus on these financially stable customers by highlighting comfort, safety and long term value. For younger and mid income customers, Sedans and Hatchbacks should be positioned as practical, reliable and cost effective choices.

- Strengthen Sedan-focused campaigns for salaried professionals:
  Salaried customers form a large and stable portion of the customer base and show a strong preference for Sedans. Campaigns aimed at this group should emphasize daily usability, comfort, fuel efficiency, and value for money, as these factors align well with their needs and purchasing behaviour.

- Position SUVs as aspirational rather than mass market products:
  SUVs appeal to a smaller but higher spending segment. Instead of promoting SUVs to all customers, marketing efforts should focus on customers with higher total household income and greater purchasing power. Messaging should highlight lifestyle benefits, premium features, and long term ownership advantages.

- Use gender based insights to refine messaging:
  Female customers, on average, spend more on vehicles. Campaigns targeted toward women can focus on premium features, comfort, safety, and design, which may resonate more strongly with this segment. Meanwhile, campaigns aimed at male customers can continue to emphasize practicality and value, especially for Sedans and Hatchbacks.

- Offer Flexible Purchase Options for Budget-Conscious Customers:
  Customers with personal or housing loans tend to be more price conscious. For this segment, Austo Motor can introduce flexible financing options, limited time discounts, or bundled service offers to reduce purchase hesitation and improve conversion rates.

- Improve customer experience through personalized communication:
  By using customer data to understand income levels, family structure, and vehicle preferences, Austo Motor can deliver more personalized communication, such as targeted emails, dealership recommendations, and customized offers, making customers feel better understood and increasing engagement.

In conclusion, to improve marketing efficiency and customer experience, Austo Motor Company should align its vehicle promotions with customer income, lifestyle and preferences. A data driven yet customer friendly segmentation strategy will help the company reach the right audience with the right message, ultimately leading to stronger sales performance and higher customer satisfaction.

10. Business Report Quality

This report is based on a structured and reliable customer dataset, covering a wide range of demographic, financial, and purchase related information. Care was taken to understand the data thoroughly before drawing conclusions, ensuring that insights are grounded in actual customer behaviour rather than assumptions.

Data quality checks were carried out at every stage, including the identification and treatment of missing values, correction of inconsistencies, and validation of unrealistic entries. Outliers were handled carefully to reduce their influence while preserving the integrity of the dataset. This ensured that the analysis reflects genuine patterns and not distortions caused by data errors.

Both univariate and bivariate analyses were used to gain a complete understanding of the data. Simple and intuitive visualizations were chosen so that findings can be easily interpreted by business stakeholders. The insights have been directly linked to the business problem and translated into practical recommendations, making the report actionable rather than purely analytical.

Overall, the report provides a balanced mix of data accuracy, clear interpretation, and business relevance, allowing decision makers to use the findings to improve marketing strategy and customer targeting.